

Overview of BST 260 Introduction to Data Science

Vince Carey, instructor, Fall 2026 edition

Prologue

The course includes activities in three main domains:

- surveying *data science and statistics* as foundations of basic integrity and efficiency of research
- introducing and using *tools for implementing data science methods* for research
- critically exploring opportunities afforded by *large language models and agentic AI* for automation of repetitive detailed tasks

A significant concern with AI adoption in education is the risk of “de-skilling”: students need to be able to

- *guide* agentic processes to accomplish clear goals
- *verify* success and supply solutions in case of agent failure
- *propose paths forward* after success is achieved

The course is intended to be highly interactive and open. Slack and GitHub are important tools facilitating visibility of our problems and solutions as they emerge over time.

Fundamental material for the course is provided in Prof. Irizarry’s text book. Please do get very familiar with it. When you think you have identified a gap in reasoning or in your understanding of aspects of the text, open a discussion either in slack or in class.

The 2026 edition of BST 260 comes at a time of heavy churn in the fields of AI, cloud computing, data storage and management. We will rely in part on materials developed by Anthony Christidis and Robert Gentleman for the 2025 edition of the course. These are all made available as legacy materials.

The course web site for 2026 will be updated frequently and I will try to make the changes as visible as possible as they are made.

Course pillars

Integrity and efficiency topics

- Files and project structures
- git, GitHub.com, project evolution
- markdown: a syntax for integrative communication

Team:

- Vince, Ethan, Gaby, Rita, Yiyang
- Check the office hours/sites on the course web page
- <https://vcitn.github.io/2026vjc>
 - this web site will be the “source of truth”, along with the canvas page, unless otherwise noted

Tools for implementing data science strategies

- R: A language and software system for statistics and data science
 - CRAN, R-universe
- Probability: models for unpredictable and uncertain phenomena
- Statistics: frameworks for measuring, predicting, interpreting *observations*

AI in relation to data science for research

- Confidentiality and compliance issues are fundamental; we will not be using any controlled-access data in this course, but the regulatory frameworks imposed by our institutions must be fully understood
- *Defining AI*: data + algorithm + interface
 - data for data science with AI: **all** open data and code are used to train foundational models
 - * code and documents used in training are of highly variable quality
 - * highly amateurish solutions are commonly encountered
 - algorithms for AI: many possibilities
 - interfaces: many possibilities
- **Always acknowledge use of AI in your coursework and research**
- Illustration: using an LLM to quiz yourself

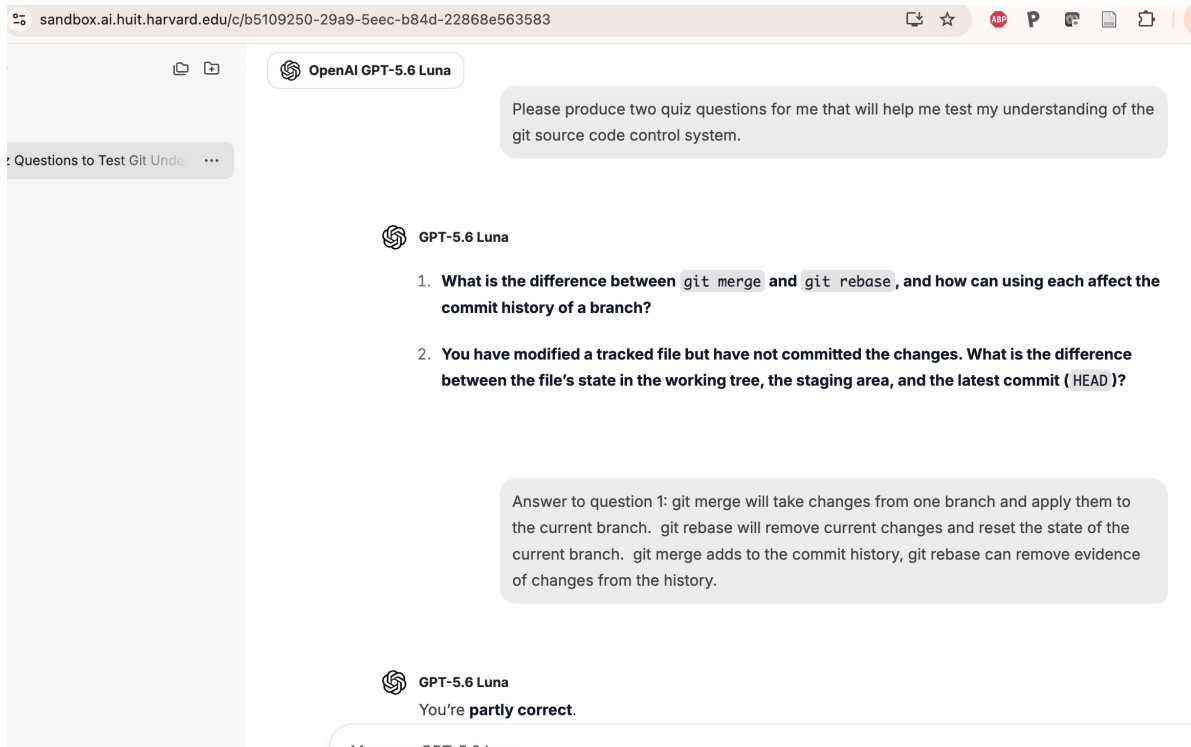


Figure 1: quiz1

sandbox.ai.huit.harvard.edu/c/b5109250-29a9-5eec-b84d-22868e563583

OpenAI GPT-5.6 Luna

Answer to question 2: the uncommitted modifications are present in the working tree. Until there is a commit, the changes are not present in the staging area. The file's state at HEAD is as prescribed in the most recent commit.

Questions to Test Git Under ...

GPT-5.6 Luna

You're **mostly correct**, but one key point needs clarification:

- **Working tree:** The files as they currently exist on disk, including modifications you have made.
- **Staging area/index:** A snapshot of the changes that will be included in the next commit. Changes can be staged **before** committing by running `git add`.
- **HEAD:** The snapshot represented by the most recent commit on the current branch.

So, uncommitted changes may exist in either—or both—the working tree and staging area:

```
text
```

```
HEAD      → last committed version
Index     → version prepared for the next commit
Working tree → current files on disk
```

Copy code

For example:

Figure 2: quiz2

It doesn't always come out well; understanding this result will be a future exercise.

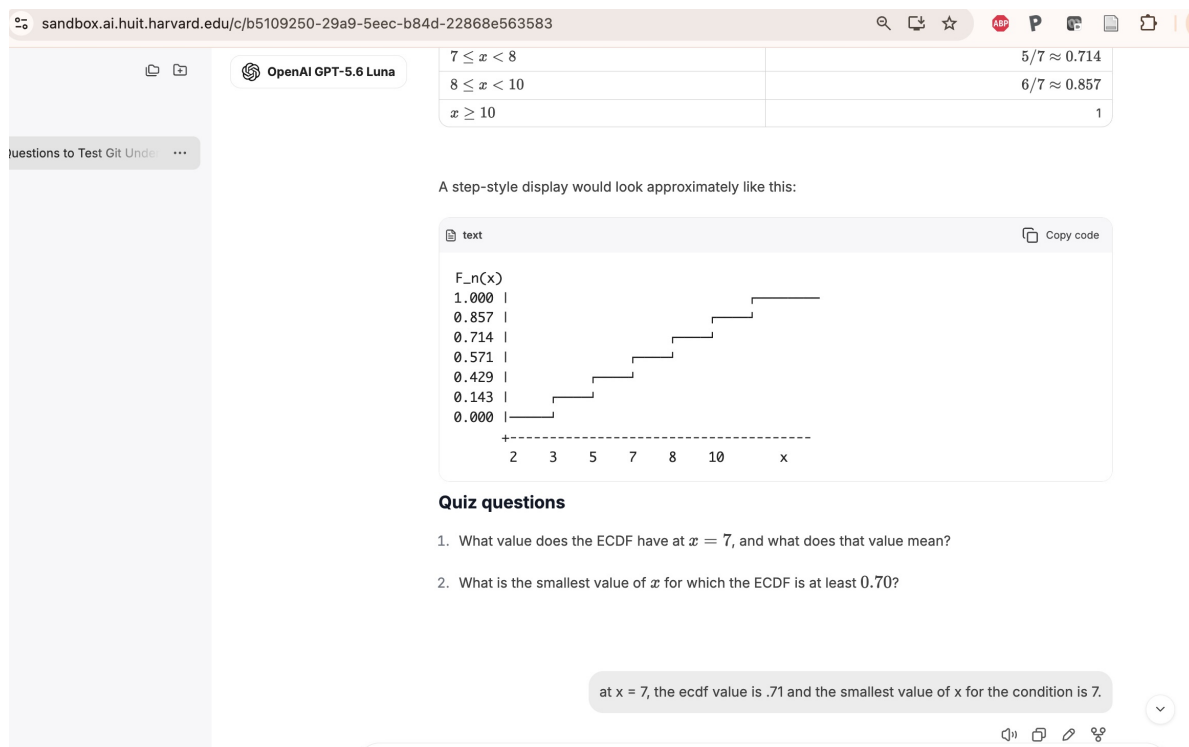


Figure 3: quiz3

Starting off

Equipment and accounts

- It is assumed you are in possession of a laptop capable of running current R and Rstudio, and on which you have administrative privileges. Please contact me if this is not so.
- You must have an account at github.com (free, use a .edu email address to get more capabilities) and at google.com (free, for working with Google Sheets, Slides, and Docs).

Forming teams during class time

- I will from time to time ask students to “count off” typically to 20 or so, so that teams of approximately 4 students each can be formed in real time
- Students will then reorganize in the classroom to form groups to collaboratively tackle a problem in real time
- Paper and pens may play a role in case you are asked to create some text or tables “by hand”

Evaluation processes; communication

- I want this course to add profoundly to your appreciation of modern data science practices, so that you see how to solve key problems of data organization, interpretation, and scientific communication that arise in your research.
- “Problem sets”, “exams/oral evals”, and “projects” are elements of evaluation that will be used.
- I would like each student to be able to prepare a web site that evolves over the course timeline, using GitHub pages
 - the site will include pages representing student encounters with data and data science methods of interest
 - the site can be public, or, if private, shared with instructor and TFs
 - this site may be the locus for work (and evaluation of work) on course problems and projects

Day 1 tasks

- install R
- install Rstudio
- use github.com: create your account
- complete google form with basic initial data
 - transform results to google sheet
 - retrieve the sheet via sheets API
 - obtain some basic statistics about course participants
- form teams of about 4 persons each via “count down”
 - create with pencil and paper a table summarizing team members’ experiences with infectious diseases and cancer
 - tables are collected and reviewed visually
 - one or more tables are turned into markdown and CSV using a multimodal LLM

Take home tasks

- create repository ‘bst260’ in your organization (account)
- invite teaching team as collaborators
- create folder ‘task-01’
- add file README.md in that folder, with a header # Overview

Last built from commit bf65a5f.